

pharmaniaga®

Ranitidine

25mg/mL Injection



COMPOSITION:

Each 2mL contains Ranitidine(as hydrochloride) 50mg.
Each 6mL contains Ranitidine(as hydrochloride) 150mg.

DESCRIPTION

Clear, colourless to yellow solution.

The compatible infusion solutions for intravenous use are as follows:

- 5% Dextrose BP
- 0.9% Sodium Chloride BP
- 0.18% Sodium Chloride and 4% Dextrose BP
- 4.2% Sodium Bicarbonate BP
- Hartmann's Solution

The solution should be clear, colourless solution when diluted.

If not required immediately, the diluted solution may be stored up to 24 hours at room temperature (<30°C) after mixing with the compatible solutions.

PHARMACODYNAMICS:

Ranitidine is a specific, rapidly acting histamine H₂-antagonist. It inhibits basal and stimulated secretion of gastric acid, reducing both the volume and the acid and pepsin content of the secretion.

Ranitidine has a relatively long duration of action and so a single 150mg oral dose effectively suppresses gastric acid secretion for 12h.

Clinical evidence has shown that oral Ranitidine combined with Amoxicillin and Metronidazole eradicates *Helicobacter pylori* in approximately 90% of patients. This combination therapy has been shown to significantly reduce duodenal ulcer recurrence. *Helicobacter pylori* infects about 95% of patients with duodenal ulcer and 80% of patients with gastric ulcer.

PHARMACOKINETICS

The bioavailability of oral Ranitidine is consistently about 50%. Peak concentrations in plasma, normally in the range 300 to 550 nanograms/mL, occur 2 to 3h after oral administration of a 150mg dose. Concentrations of Ranitidine in plasma are proportional to oral dose up to and including 300mg. Absorption of Ranitidine after i.m. injection is rapid and peak plasma concentrations are usually achieved within 15min of administration.

Ranitidine is not extensively metabolised. The metabolism of Ranitidine is similar after both oral and i.v. dosing; about 6% of the dose being excreted in urine as the N-oxide, 2% as the S-oxide, 2% as desmethylranitidine and 1 to 2% as the furoic acid analogue.

In balance studies with 150mg 3H-Ranitidine 93% of an i.v. dose was excreted in urine and 5% in faeces; 60 to 70% of an oral dose was excreted in urine and 26% in faeces. Analysis of urine excreted in the first 24h after dosing showed that 70% of the i.v. dose and 35% of the oral dose were eliminated unchanged. Elimination of the drug is primarily by tubular secretion. The elimination half-life is 2 to 3h.

INDICATIONS:

- Duodenal Ulcer.
- Benign Gastric Ulcer.
- Post-operative Ulcer.
- Reflux Oesophagitis.
- Zollinger-Ellison Syndrome.
- Prophylaxis of stress ulceration in seriously ill.
- Prophylaxis of recurrent haemorrhage from peptic ulcer.
- Prophylaxis of Mendelson's syndrome.

RECOMMENDED DOSAGE:

Ranitidine Injection may be given as:-

- a slow (over 2 min) i.v. injection of 50mg, diluted to a volume of 20mL, every 6 to 8h.
- an intermittent i.v. infusion at 25mg/h for 2h, repeated at 6 to 8h intervals.
- an i.m. injection of 50mg every 6 to 8h.

Prophylaxis of mendelson's syndrome

For prophylaxis of Mendelson's syndrome 50mg by i.m. or slow i.v. injection 45 to 60mins before induction of general anaesthesia.

Prophylaxis of haemorrhage from stress ulceration in seriously ill patients or prophylaxis of recurrent haemorrhage in patients bleeding from peptic ulceration

In the prophylaxis of haemorrhage from stress ulceration in seriously ill patients or prophylaxis of

recurrent haemorrhage in patients bleeding from peptic ulceration parenteral administration may be continued until oral feeding commences. Patients considered to be still at risk may then be treated with Ranitidine tablets 150 mg twice daily.

In the prophylaxis of upper gastrointestinal haemorrhage from stress ulceration in seriously ill patients a priming dose of 50mg as a slow intravenous injection followed by a continuous intravenous infusion of 0.125-0.250mg/kg/h may be preferred.

Children

Use of Ranitidine injection in children has not been evaluated

Renal Impairment

Accumulation of Ranitidine with resulting elevated plasma concentrations will occur in patients with severe renal impairment (creatinine clearance less than 50 mL/min). Ranitidine injection is recommended to be administered in doses of 25mg.

Compatibilities

Ranitidine Injection has been shown to be compatible with the following intravenous infusion fluids:

- 0.9% Sodium chloride BP
- 5% Dextrose BP
- 0.18% Sodium chloride and 4% Dextrose BP
- 4.2% Sodium bicarbonate BP
- Hartmann's solution.

Route of Administration

Parenteral

- Intravenous injection (slow IV injection and intermittent IV infusion)
- Intramuscular injection

CONTRAINDICATIONS:

Contraindicated in patients known to have hypersensitivity to any component of the preparation.

WARNINGS AND PRECAUTIONS:

The possibility of malignancy should be excluded before commencement of therapy in patients with gastric ulcer and patients of middle age and over with new or recently changed dyspeptic symptoms, as treatment with Ranitidine may mask symptoms of gastric carcinoma.

Ranitidine is excreted via the kidney and so plasma levels of the drug are increased in patients with severe renal impairment. The dosage should be adjusted as detailed above under Recommended Dosage in Renal Impairment.

Rare clinical reports suggest that Ranitidine may precipitate acute porphyric attacks. Ranitidine should therefore be avoided in patients with a history of acute porphyria. Regular supervision of patients who are taking non-steroidal anti-inflammatory drugs concomitantly with oral Ranitidine is recommended, especially in the elderly and in those with a history of peptic ulcer.

In patients such as the elderly, persons with chronic lung disease, diabetes or the immunocompromised, there may be an increased risk of developing community acquired pneumonia. There is an increased risk of developing community acquired pneumonia in current users of H₂ receptor antagonists.

Bradycardia in association with rapid administration of Ranitidine injection has been reported rarely, usually in patients with factors predisposing to cardiac rhythm disturbances. Recommended rates of administration should not be exceeded.

The use of higher than recommended doses of intravenous H₂- antagonists has been associated with rises in liver enzymes when treatment has been extended beyond five days.

Effects on Ability to Drive and Use Machines

None reported.

INTERACTIONS WITH OTHER MEDICAMENTS:

Ranitidine has the potential to affect the absorption, metabolism or renal excretion of other drugs. The altered pharmacokinetics may necessitate dosage adjustment of the affected drug or discontinuation of treatment. Interactions occur by several mechanisms including:

Inhibition of cytochrome P450-linked mixed function oxygenase system:

Ranitidine at usual therapeutic doses does not potentiate the actions of drugs which are inactivated by this enzyme system such as Diazepam, Lidocaine, Phenytoin, Propranolol and Theophylline. There have been reports of altered prothrombin time with coumarin anticoagulants (e.g. Warfarin). Due to the narrow therapeutic index, close monitoring of increased or decreased prothrombin time is recommended during concurrent treatment with Ranitidine.

Competition for renal tubular secretion:

Since Ranitidine is partially eliminated by the cationic system, it may affect the clearance of other drugs eliminated by this route. High doses of Ranitidine (e.g. such as those used in the treatment of Zollinger-Ellison syndrome) may reduce the excretion of procainamide and N-acetylprocainamide resulting in increased plasma levels of these drugs.

Alteration of gastric pH

The bioavailability of certain drugs may be affected. This can result in either an increase in absorption (Triazolam , Midazolam, Glipizide) or a decrease in absorption (e.g. Ketoconazole, Atazanavir, Delaviridine, Gefitinib).

There is no evidence of an interaction between oral Ranitidine and Amoxicillin and Metronidazole.

If high doses (2g) of Sucralfate are co-administered with oral Ranitidine the absorption of the latter may be reduced. This effect is not seen if Sucralfate is taken after an interval of 2h.

Incompatibilities:

Although compatibility studies have only been undertaken in polyvinyl chloride infusion bags (in glass for Sodium bicarbonate BP) and a polyvinyl chloride administration set it is considered that adequate stability would be conferred by the use of a polyethylene infusion bag.

PREGNANCY AND LACTATION:

Ranitidine crosses the placenta and is excreted in breast milk. Like other drugs Ranitidine should only be used during pregnancy or during nursing if considered essential.

SIDE EFFECTS:

Blood & Lymphatic System Disorders

Very Rare: Blood count changes (leucopenia, thrombocytopenia). These are usually reversible. Agranulocytosis or pancytopenia, sometimes with marrow hypoplasia or marrow aplasia.

Immune System Disorders

Rare: Hypersensitivity reactions (urticaria, angioneurotic oedema, fever, bronchospasm, hypotension and chest pain).

Very Rare: Anaphylactic shock

These events have been reported after a single dose.

Psychiatric Disorders

Very Rare: Reversible mental confusion, depression and hallucinations.

These have been reported predominantly in severely ill and elderly patients.

Nervous System Disorders

Very Rare: Headache (sometimes severe), dizziness and reversible involuntary movement disorders

Eye Disorders

Very Rare: Reversible blurred vision.

There have been reports of blurred vision, which is suggestive of a change in accommodation.

Cardiac Disorders

Very Rare: As with other H2 receptor antagonists bradycardia, A-V block and, with the injection only, asystole.

Vascular Disorders

Very Rare: Vasculitis

Gastrointestinal Disorders

Very Rare: Acute pancreatitis, diarrhoea.

Hepatobiliary Disorders

Rare: Transient and reversible changes in liver function tests.

Very Rare: Hepatitis (hepatocellular, hepatocanalicular or mixed) with or without jaundice, these were usually reversible.

Skin and Subcutaneous Tissue Disorders

Rare: Skin rash.

Very Rare: Erythema multiforme, alopecia .

Musculoskeletal and Connective Tissue Disorders

Very Rare: Musculoskeletal symptoms such as arthralgia and myalgia.

Renal and Urinary Disorders

Very rare: Acute interstitial nephritis.

Reproductive System and Breast Disorders

Very Rare: Reversible impotence, breast symptoms and breast conditions (such as gynaecomastia and galactorrhoea).

OVERDOSE TREATMENT:

Ranitidine is very specific in action and no particular problems are expected following overdosage with ranitidine formulations. Symptomatic and supportive therapy should be given as appropriate.

STORAGE CONDITIONS:

Store below 30°C.

The solution should be clear, colourless solution when diluted. If not required immediately, the diluted solution may be stored up to 24 hours at room temperature (<30°C). after mixing with the compatible solutions. Discard unused portion.

SHELF LIFE

3 years

The solution should be clear, colourless solution when diluted. If not required immediately, the diluted solution may be stored up to 24 hours at room temperature (<30°C). after mixing with the compatible solutions. Discard unused portion.

DOSAGE FORMS AND PACKAGING AVAILABLE:

10 X 2mL ampoule (clear)

10 X 6mL vial (clear)

PRODUCT REGISTRATION HOLDER/MANUFACTURER

PHARMANIAGA LIFESCIENCE SDN BHD (82685-T)

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